

Beyond Clean Demonstrations: Coverage-Aware Force-Torque Data Collection for Contact-Rich Manipulation

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Abstract—Imitation learning for contact-rich manipulation is often limited not by policy architecture, but by the quality and structure of the demonstration dataset. We report practical lessons learned while developing a real-robot peg-in-hole insertion system using force-torque (FT) augmented demonstrations. Our experiments indicate that clean demonstrations—where the operator completes the task in a single smooth motion—are insufficient even when FT observations are added to the policy input, as the dataset lacks coverage of the contact configurations encountered at deployment. Augmenting the dataset with recovery demonstrations can substantially improve performance, but only when demonstrations are structured appropriately, as naively segmenting contact-retract cycles into independent episodes causes the policy to become trapped in a repetitive loop in most trials, executing recovery behaviors without completing insertion. Our central finding is that, on this task, recovery demonstrations were most effective when *continuously connected* to successful task completion, allowing the policy to retract from contact and reattempt insertion, succeeding in most trials. We additionally present a real-time contact distribution visualization interface that enables coverage-aware demonstration collection. These findings highlight that how demonstrations are structured and collected matters as much as how many are gathered.

I. INTRODUCTION

Imitation learning has emerged as a powerful paradigm for robot manipulation, enabling policies to acquire complex skills directly from human demonstrations without manually designed reward functions [1, 2]. A common intuition is that more demonstrations yield better policies, yet adding more demonstrations of the same type often gives diminishing returns, while a small number of well-chosen ones can substantially improve performance [3, 4]. This raises a practical question: *how should demonstrations be collected and structured to maximize their value?*

The challenge is particularly acute in contact-rich manipulation, especially when the object is handled with a dexterous robot hand. Tasks such as peg-in-hole insertion, where small variations in the grasp readily introduce discrepancy, demand precise resolution of contact states that are largely inaccessible from an egocentric RGB camera: contact onset, force direction, and misalignment all occur at the physical interaction point, where pixel intensities carry little discriminative information. Humans handle this naturally through haptic perception. When inserting a peg under occlusion, one relies not on visual confirmation but on the feel of resistance, the sense of angular misalignment, and the haptic feedback of

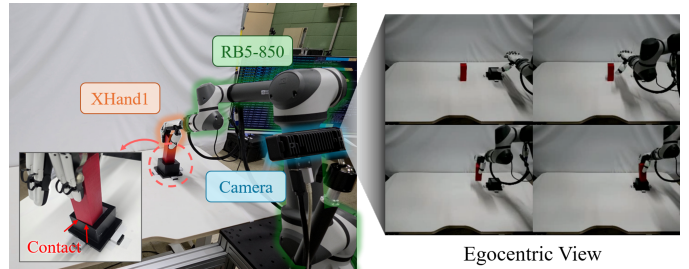


Fig. 1. Real-robot peg-in-hole insertion task. A rectangular peg is grasped with a dexterous multi-fingered hand and inserted into a matching hole. Because the hand grasps the peg with slight pose variation, the peg often meets the hole boundary rather than entering cleanly. This critical contact phase occurs at the hole entry, which is partially occluded from the egocentric camera.

successful seating. Robots equipped with only RGB cameras largely lack this capability.

Force-torque (FT) sensors offer a natural path toward closing this gap, and recent work has begun incorporating FT information into imitation learning policies for contact-rich tasks [5, 6]. Yet instrumentation alone is insufficient. If the demonstration dataset consists of clean executions in which the operator inserts the peg as directly as possible, the training data will contain few examples of the interactions FT sensing is meant to capture: entry contact, recovery, and force-guided correction. The sensor records the contact, but the dataset never shows the policy how to respond to it.

In this work, we document our experience building an FT-augmented imitation learning system on a real-robot peg-in-hole insertion task, in which a dexterous hand grasps and inserts the peg. To collect recovery demonstrations with diverse contact configurations, we developed a real-time visualization interface that displays the spatial distribution of contact events during collection. Using this setup, we collected recovery demonstrations and augmented our training dataset, a process from which a key finding emerged: naively segmenting contact-retract cycles into independent episodes causes the policy to become trapped in a repetitive loop in most trials, executing recovery behaviors without completing insertion. Recovery demonstrations should be *continuously connected* to the successful completion of the task, as this connection allows the policy to learn to recover in order to complete the task.

II. SYSTEM AND TASK DESCRIPTION

Our system consists of a 6-DOF robotic arm (Rainbow Robotics RB5-850) with a dexterous hand (Robotera XHand1), a wrist-mounted six-axis force-torque (FT) sensor, and a fixed egocentric RGB camera, mounted to approximate a human operator’s head-level viewpoint. The entire system runs on ROS2. Demonstrations are collected via teleoperation with synchronized recording of RGB images, joint states, end-effector poses, actions, and FT measurements $\mathcal{F} = [f, \tau] \in \mathbb{R}^6$.

The task is peg-in-hole insertion: a rectangular prismatic object must be grasped with the robot hand and inserted into a matching hole (Fig. 1). At deployment, even small positional deviations cause the peg to contact the hole boundary rather than enter it. We refer to this as an *entry contact* event: the resulting force response signals that misalignment has occurred, information that is not recoverable from visual observations alone, as a misaligned peg looks nearly identical to a correctly positioned one from an egocentric camera. Demonstrations collected without deliberate contact-recovery behavior, where the operator simply attempts to insert the peg as directly as possible, contain almost no entry contact events. Policies trained on such data consequently have no basis for responding when entry contact occurs at deployment. The following sections describe how we addressed this gap through targeted demonstration collection and iterative experimentation.

Throughout all experiments, we fix the policy architecture: frozen DINOv2 [7] (ViT-S/14) with a fine-tuned MLP projection head; FT and proprioceptive measurements concatenated with the visual embedding; and a Diffusion Policy [1] head with action chunking. Keeping the architecture constant allows performance differences to be primarily attributed to the demonstration composition rather than the modeling capacity.

III. DEMONSTRATION COLLECTION FRAMEWORK

A. Data Collection Protocol

We collect demonstrations in two stages. **Baseline (25 demos)**: the operator inserts the peg in a single smooth motion from a fixed start position, minimizing unnecessary contact. We refer to these as *clean demonstrations*: collected with the primary goal of task completion, they contain limited entry contact coverage. **Recovery (20 demos)**: the operator deliberately contacts the hole boundary several times before completing insertion, guided by the real-time visualization interface described below. Each demonstration contains multiple contact-retract cycles followed by a single successful insertion. These demonstrations are subsequently segmented into training episodes under two different strategies, described in Section III-C.

B. Real-Time Contact Distribution Visualization

To collect recovery demonstrations that cover a diverse range of contact configurations, rather than repeatedly sampling the same region, the operator needs feedback on what the current dataset already covers. Without such feedback, there is no direct way to assess whether newly collected

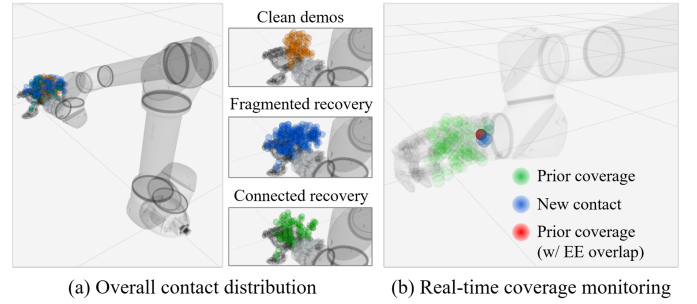


Fig. 2. (a) Overall contact distribution across three demonstration conditions: clean demos (orange), fragmented recovery (blue), and connected recovery (green). (b) Real-time coverage monitoring interface: prior coverage (green), new contact detection $\|f\| > f_{th}$ (blue), and prior coverage w/ EE overlap (red), which prompts the operator to redirect collection toward uncovered regions.

demonstrations add coverage or simply duplicate existing data. We develop a real-time interface that renders end-effector positions at contact events, detected from force magnitude ($\|f\| = \sqrt{f_x^2 + f_y^2 + f_z^2} > f_{th}$) as spherical markers in 3D, updating as each new demonstration is added. Contact points from previously collected demonstrations are rendered in green; newly detected contact points in the current session appear in blue and transition to green as collection continues. When the current end-effector position enters the vicinity of an existing contact marker, that marker changes to red, implicitly signaling the operator to collect data elsewhere. Fig. 2(a) shows the contact distribution across the three demonstration conditions described below, and Fig. 2(b) illustrates the real-time monitoring interface.

C. Episode Segmentation

Once recovery demonstrations are collected, the structure of the resulting training episodes has a significant impact on policy performance. Our initial assumption was that providing the policy with examples of contact-retract behavior would be sufficient for it to learn to recover from entry contact. Under this assumption, we segmented each contact-retract cycle within the 20 recovery demonstrations into an independent episode, yielding 92 short recovery segments (**fragmented strategy**). However, as we describe in Section IV, this led to an unexpected failure mode. We therefore re-segmented the same 20 demonstrations under an alternative strategy: rather than storing the full multi-contact sequence, we manually identified the final one or two contact-retract cycles before successful insertion and extracted each as a single training episode (**connected strategy**).

IV. EXPERIMENTAL RESULTS

We evaluate three demonstration conditions, each trained with the same policy architecture and evaluated over 10 insertion attempts per condition. Trials in which grasping failed are excluded and repeated, as our focus is on policy behavior during the insertion phase.

Clean demonstrations (vision only and vision + FT). The vision-only policy learned a reasonable approach trajectory but

consistently failed at the insertion phase. Any small positional deviation caused the peg to press against the hole boundary; without a learned response to this contact, the policy continued applying downward force, accumulating contact forces that posed a hardware risk (0/10). Adding FT observations to the same dataset did not substantially resolve the failure. The policy occasionally reacted to contact forces, but lacked consistent recovery behavior, as the training data contained limited entry contact coverage (0/10). Together, these outcomes point to demonstration coverage, rather than observation modality, as the bottleneck. FT observations let the policy detect contact, but without demonstrations of how to respond, there is no behavior for that signal to inform.

Fragmented recovery augmentation. To provide the policy with entry contact experience, we augmented the clean demonstrations with 92 fragmented recovery segments and trained with vision + FT observations. Unexpectedly, this degraded performance below the clean-only baseline (1/10). At deployment, the policy entered a repetitive contact-retract loop in most trials, completing insertion in only 1 of 10 attempts. Because a diffusion policy makes retraction the modal response to contact when short contact-retract episodes dominate the dataset, the policy retracts reliably but has no learned path from recovery back to insertion, because the two were never connected in the training data.

Connected recovery demonstrations. We re-segmented the same 20 recovery demonstrations under the connected strategy: for each demonstration, we extracted the final one or two contact-retract cycles before successful insertion as a single episode. We retrained on this dataset combined with the 25 clean demos, achieving 7/10 insertions, a substantial improvement over all prior conditions. The policy now responded to entry contact by retracting and reattempting insertion, rather than continuing to press or becoming trapped in a contact-retract loop. The results suggest that recovery demonstrations should not be isolated from their outcome: the policy needs to observe recovery and task completion together to learn that one leads to the other. Segmenting recovery from its outcome leaves the policy without a path from recovery back to task completion.

TABLE I
INSERTION SUCCESS RATES ACROSS DEMONSTRATION CONDITIONS (10 TRIALS PER CONDITION).

Condition	Observation	Success
Clean demos	Vision only	0/10
Clean demos	Vision + FT	0/10
Fragmented recovery	Vision + FT	1/10
Connected recovery	Vision + FT	7/10

V. DISCUSSION AND FUTURE WORK

Our motivation for fragmented segmentation was to prevent the policy from learning that multiple contact attempts are a necessary part of the task. By isolating each contact-retract cycle as an independent episode, we hoped to expose the

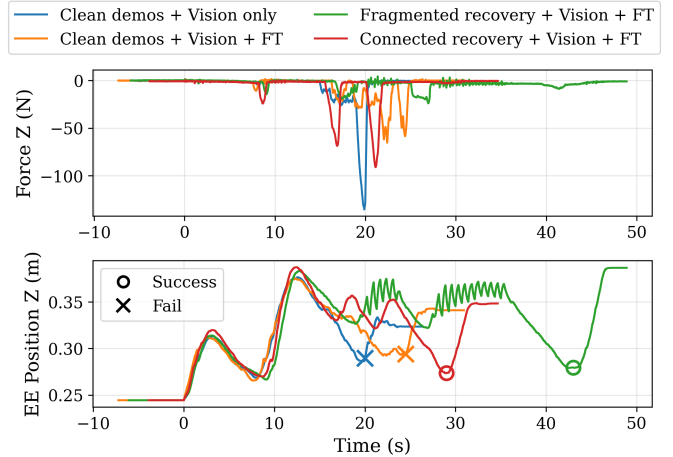


Fig. 3. Representative Z-axis force (top) and end-effector Z-axis position (bottom) at deployment for each condition. Clean demonstrations (vision only) show continuous force accumulation at contact with no recovery response. Clean demonstrations (vision + FT) show partial reactivity to contact but fail to complete insertion. Fragmented recovery shows irregular contact-retract behavior before eventual insertion. Connected recovery shows a contact event followed by retraction and successful insertion.

policy to recovery behavior without encoding the full multi-contact sequence. This reasoning is intuitive, yet our results show that it severely degrades performance. Because a diffusion policy learns a distribution over actions conditioned on the current observation [1], short contact-retract episodes become a dominant behavioral mode, trapping the policy in a loop. The critical factor is not episode quantity but semantic completeness. Each episode should preserve the causal association between recovery behavior and task completion. We hypothesize that this principle may extend beyond peg-in-hole insertion to any contact-rich task in which recovery behavior is a necessary component of task completion, though empirical validation across tasks and policy architectures remains for future work.

The real-time visualization interface introduced in Section III represents a step toward more deliberate, coverage-aware demonstration collection. By making the spatial distribution of contact events visible during collection, it creates a feedback loop between dataset state and operator behavior. A natural extension would formalize this notion of coverage to identify sparse regions in the contact distribution, and quantify recovery diversity via distribution distance measures. We leave empirical validation of such metrics to future work, alongside examining whether the connected segmentation principle generalizes across policy architectures beyond diffusion-based approaches.

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